



Fig. 12: DDRAM

generator ROM) and CGRAM (character generator RAM).

DDRAM. DDRAM is a sort of RAM where data can be written to and read from, but loses its contents when the power is removed. It is used to store the characters (up to 80) that need to be displayed (see Fig. 12).

If the cursor is on, it is displayed at the location that is currently addressed. When a character appears at the cursor position, it automatically moves to the next addressed location.

CGROM. It contains the default character map with all the characters that can be displayed on the screen. Each character is assigned to one memory location. Addresses of CGROM locations match the characters of ASCII. For example, if the program currently being executed encounters a command "send character P to port," binary value 0101 0000 appears on the port. This value is ASCII equivalent of character P which will be displayed on the LCD. This applies to all the letters of the alphabet (capital and small), but not to the numbers. In order to display the numbers correctly, it is necessary to add decimal number 48 to each of the values prior to sending them to the LCD.

For details of the LCD memory, refer to the datasheet of HD44780 controller included in the EFY-CD.

LCD connections. Depending on the number of lines used for connecting an LCD to the microcontroller, there are 8-bit and 4-bit modes of LCD connection. The appropriate mode is selected at the beginning of the operation in a process called 'initialisation.' 8-bit LCD mode uses D0 through D7 output lines to transfer the data. In 4-bit mode, only four higher bits (D4 through D7) are used for communication, while others may be left unused.

Each data is sent to the LCD in two steps: four higher bits are sent first (normally via D4 through D7 lines), followed by four lower bits. Initialisation enables the LCD to link and interpret the received bits correctly. In our development board, we have used 4-bit mode of LCD connection. The main purpose of 4-bit LCD mode is to save the valuable I/O pins of the microcontroller.

LCD initialisation. The content of the LCD memory is automatically cleared when it is powered up. It takes approximately 15 ms to clear the memory data. After that, the display is ready to operate.

Usually, automatic reset is done without any problem. If for any reason the power supply voltage does not reach its full value within 10 ms, the display starts performing completely unpredictably.

The LCD is initialised through software instructions. Typically, driver files for the LCD module consist of the `lcd.h` header file and the accompanying `lcd.c` source file. Once you include driver files for the LCD module, all that you need to know in order to operate the module are a few user-defined functions as given below:

```
extern void lcd_init(uint8_t
dispAttr);
Initialize display and select type
of cursor.
extern void lcd_clrscr(void);
Clear display and set cursor to home
position.
extern void lcd_home(void);
Set cursor to home position.
extern void lcd_gotoxy(uint8_t x,
uint8_t y);
Set cursor to specified position.
extern void lcd_putc(char c);
Display character at current cursor
position.
```

Are You In Suspense?

Are you wondering what happened to your order or payment, even after a fortnight? But did you mention your name and address clearly while sending payment, and what the payment was for?

We get several payments every month without adequate instructions or even identity of the person sending the payment. As a result we are unable to take appropriate action on these, till we get a complaint - if at all.

At EFY, we want you to be our satisfied customer. So please do let us know in case your payment or instructions are not even acknowledged within a fortnight. We are not that lazy!

Email: support@efyindia.com
Ph: 011-26810601 / 02 / 03 extn: 242